Irving F. Sanchez

U.S. Air Force Veteran | CS Graduate | Software Engineering

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SUMMARY

Computer Science graduate with a minor in Mathematics. Leveraging 12+ years of hands-on experience across electrical systems, PLCs, industrial automation, robotics, and mission-critical infrastructure. U.S. Air Force veteran bringing a rare combination of field-tested electrical troubleshooting of complex hardware, low-voltage electronics, fire alarm systems, power distribution systems, automation equipment, and software-adjacent industrial control environments. Skilled in software development, full-stack web technologies, embedded systems concepts, cloud deployment, and Agile/Scrum collaboration.

Education

B.S. in Computer Science—Minor in Mathematics

Lewis University, Romeoville, IL — 2019 - 2025

Focus on Software Development, Machine Learning, and Computer Engineering.

Certification—Journeyman Electrician (3E051)

United States Air Force, Wichita Falls, TX — 2010 - 2023

Trained in low-high voltage electrical systems, residential, industrial, and airfields.

Technical Projects

Syllabye—Senior Capstone Project

Lewis University — 2023 – 2025 — Romeoville, IL

Technologies: React, Vite, JS, HTML5, CSS, Tailwind CSS, Bootstrap, Firebase Azure Web Apps, Google Cloud, Git/GitHub, react-i18next, NPM, Babel, VS-Code

Architected and deployed a full-stack, componentized web application to streamline academic syllabus creation, access and management. (Users 1-171)

- Product Owner | Front-End & System Architecture | Documentation
- Engineered real-time bilingual language switching (English/Spanish) utilizing react-i18next, ensuring broad accessibility and seamless UX.
- Integrated Firebase for scalable PDF storage and deployed the production environment via Azure Static Web Apps and Google Cloud.
- Led Agile/Scrum workflows as Product Owner, coordinating sprints, managing the product backlog, and ensuring successful handoff of a modular, sustainable codebase, and GitHub-based version control.
- Documented project structure, developer setup, i18n guidance, and future enhancement recommendations to support maintainability.

Telemetry IoT Dashboard & L.E.A.H. Automation Cell

Personal Project — 2025 – 2026 — Willowbrook, IL

Technologies: Arduino UNO Q, C++/Python, React, REST API, SQL/Firebase, Sensors, Azure, Linux, Git/GitHub, ESP32, HuskyLens

Engineered and built the Learning Embedded Autonomous Hardware (L.E.A.H.). A closed-loop mini-industrial automation cell with a 6-axis robotic-arm & telemetry.

- Built & programmed the robotic automation cell powered by an ESP32 controller and HuskyLens AI vision to identify items, sort, and coordinate commands with the safety hardware/software.
- Integrated an Arduino UNO Q and peripherals as a safety PLC for hardware interlocks & incorporated manual safety e-stops.
- Architected a React/Node.js Telemetry IoT Dashboard to monitor live machine-state data and system faults to mimic real life automation.

STRENGTHS

Problem-Solving

Bridged hardware-software systems in electrical and automation roles. Resolved complex electrical/software-adjacent issues, including troubleshooting airfield grids and debugging assembly-line automation systems.

Collaboration

Led and trained a team of electricians to deploy military mission essential electrical systems. Partnered with software developers to integrate PLCs and robotics, ensuring seamless automation.

Work Ethic

Maintained mission-critical systems under tight deadlines with precision, reliability, and accountability. Continues to build advanced skills in software engineering, embedded systems, and computer engineering concepts through project-based development.

Technical Skills

Programming Languages: Python, Java, C++, C#, C, PHP MATLAB, HTML, CSS, JavaScript, SQL

Frameworks & Web Technologies: React, Vite, Node.js Express, .NET, REST APIs, WebSockets, Tailwind CSS Bootstrap

Tools & Technologies

Embedded/Automation: Arduino UNO Q, Sensors, PLC Concepts, Ladder Diagrams, Function Block Diagrams Circuit Analysis, Raspberry Pi

Tools & Platforms: Git, GitHub, VS Code, Linux, WSL Docker, Azure, Azure Static Web Apps, Google Cloud Firebase, VirtualBox, VMware, Hyper-V, LTspice Simulink, AutoCAD, MS Office

Professional Skills

Related Skills: Agile/Scrum, Product Ownership, Blue-Print and Schematic Interpretation, Technical Documentation, Bilingual English/Spanish

EXPERIENCE

Journeyman Electrician

United States Air Force 2010 – 2023 Peru, IN

Responsible for troubleshooting and maintaining low and high voltage systems.

- Maintained substation and underground power distribution systems.
- Wired electrical components for residential and industrial facilities.
- Led airmen in electrical projects provided by shop foreman.
- Certified bucket truck driver, and in demolition class vehicles.
- Read and interpreted wiring diagrams for electrical assembly.
- Ensured maintenance of airfield lighting and electrical grids.
- Diagnosed and maintained fire alarm systems, low-voltage electronics and related electrical components to support facility safety.
- Attained Journeyman Level Certification and Security Clearance.

Electrician

Flow Pro Inc. 2019 - 2019 Romeoville, IL

Constructed and wired electrical control panels and high-voltage relay grids.

- Analyzed and interpreted wire diagrams for magnetic filtration systems.
- Collaborated with lead electrical engineer for quality control and assisted in Ladder Diagram design.
- Owned and maintained proper electrical tools and PPE.
- Constructed electrical panel enclosures for power distribution/management.

Electrician (Contractual)

BW Container System 2018 – 2019 Bolingbrook, IL

Assembled electrical panels for industrial food and beverage conveyor lines.

- Wired PLC and classic relay systems for food automation packaging.
- Read and interpreted ladder diagrams and electrical schematics.
- Troubleshot electrical systems and coordinated with a team of electrical and mechanical engineers for quality assurance.

Electrician (Contractual)

Felsomat USA Inc. 2017 - 2018 Schaumburg, IL

Built and integrated precision laser-cutting automated assembly systems for industrial manufacturing customers.

- Assembled and wired electrical control panels and high voltage relay grids.
- Wired PLCs (Programmable Logic Controller) and related peripherals.
- Collaborated with controls engineer to ensure proper function of all electrical components using ladder logic and/or function block diagram.
- Wired and tested programmable robotics for assembly line automation.

Electrician & Mechanic (Contractual)

Messina Group 2017 – 2017 Logan Square, IL

Assembled industrial washer and dryer machines.

- Worked on industrial machine tool and dye equipment.
- Wired multi-panel electrical control enclosures.
- Read and interpret wire diagrams for electrical assembly.

Electrician (Contractual)

Thyssenkrupp Engineering 2016 – 2017 Auburn Hills, MI

Assembled and wired robotic automobile automation lines for (GM, Ford, Chrysler)

- Wired automobile electrical-mechanical automation systems, power distribution panels, HMIs, assembly robotics and emergency safety systems.
- Wired PLCs (Programmable Logic Controller) and related peripherals.
- Collaborated with automation engineers ensuring seamless functionality of automation systems and robotics using ladder logic and C++.
- Troubleshot electrical and mechanical components to ensure proper and secure installation.